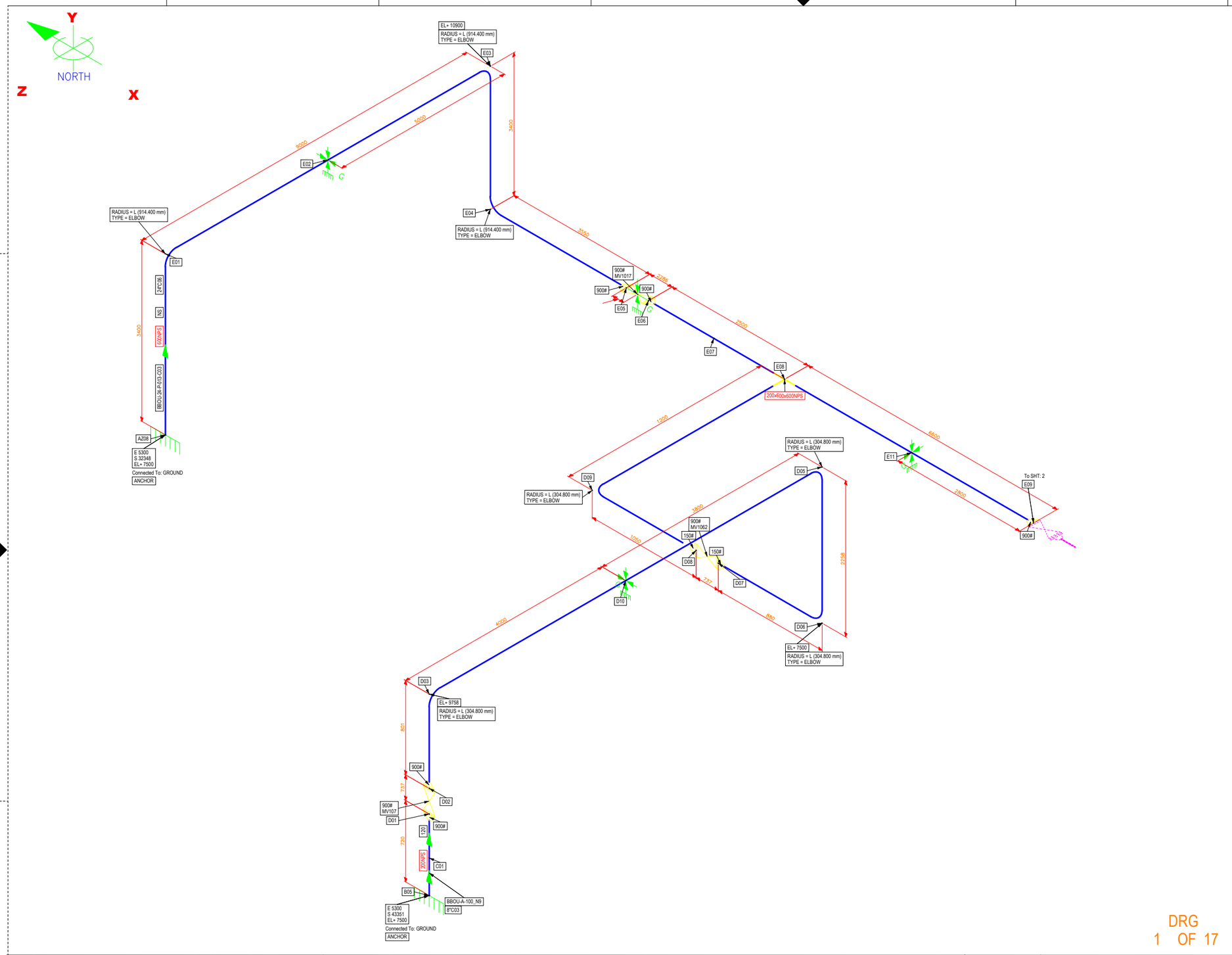

DRG
1 OF 17

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MODEL INFO                    AutoPIPE Ver: 12.06.00.48
System: BBOU. _ASME B31.3
Piping Code:   ASME B31.3 - 2016
Model Rev: 25          Ambient: 24.0 (deg C )
```

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"C06	API5L-X60	600.00	NS	609.60	38.890	6.000	4.861		0.700
8"C03	API5L-B	200.00	120	219.07	18.263	3.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
D01	200.00	CATALOG	900#	578.79
D07	200.00	CATALOG	900#	578.79
E05	600.00	CATALOG	900#	6785.86

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
D01	200.00	WELDNECK	900#	168.74	84.37
D02	200.00	WELDNECK	900#	168.74	84.37
D07	200.00	WELDNECK	150#	39.01	
D08	200.00	WELDNECK	150#	39.01	
E05	600.00	WELDNECK	900#	1764.00	
E06	600.00	WELDNECK	900#	1764.00	
E09	600.00	WELDNECK	900#	1764.00	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
D10 1	Guide-H	Ground		3.00	3.00	0.10
D10 1	Guide-V	Ground			3.00	0.10
E02 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
E02 1	Guide-V	Ground	Rigid		3.00	0.10
E05 M1	V-stop	Ground	Rigid		2540.00	0.10
E11 1	Guide-H	Ground		3.00	3.00	0.10
E11 1	Guide-V	Ground			3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
D10 1				1.32	-6.60	-5.24	GT1P1(1)
D10 1				5.98	-8.60	1.45	GT2P2(1)
D10 1				6.21	-8.59	1.70	GT2(1)
D10 1				4.33	-7.60	-0.96	Gravity(1)
E02 1	24335.65	-87608.8	10595.38	3.43		2.82	Thermal 2(1)
E02 1	24335.65	-146253	10595.38	3.00		2.85	GT2(1)
E02 1	-79216.1	4444.86	-6556.81	-3.00	2.66	-3.89	GT1P1(1)
E02 1		-58644.2		-0.43		0.03	Gravity(1)
E05 M1	-15203.8	-215060	-15208.8	-11.73		-11.63	GT1P1(1)
E05 M1	-20628.6	-215930	-6366.96	-0.83		-0.10	GT3P3(2)
E05 M1	-41040.3	-11683.9	-7073.15	-0.95		-0.26	Pressure 3(2)
E05 M1		-189165				0.11	Gravity(1)
E11 1				3.02	-0.55	6.58	GT2(1)
E11 1				-4.70	-0.26	-10.89	GT1P1(1)
E11 1					-0.24	-0.08	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and
movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AZ08	504.89	-11683.1	12144.22	20491.09	9019.47	-4273.81	Pressure 3(2)
AZ08	433.61	37586.09	-17324.5	14952.92	-64641.1	22859.25	GT2(1)
AZ08	1297.69	54065.79	-16875.3	13175.67	-70101.6	12987.72	Thermal 2(1)
AZ08	25235.94	-52632.3	14076.10	13938.30	72320.02	-126899	GT1P1(1)
B05	-0.16	-17630.3	-801.03	-10325.8	-369.40	-3369.28	GT1P1(1)
B05	1118.75	-17115.6	2543.12	-2855.63	-4403.36	-5089.13	GT2(1)
B05	758.37	-18186.5	1432.90	-6115.54	-3136.43	-4863.75	GT3(2)
B05	1287.18	-17313.4	1271.48	-5695.89	-3281.97	-5648.17	GW1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap — Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

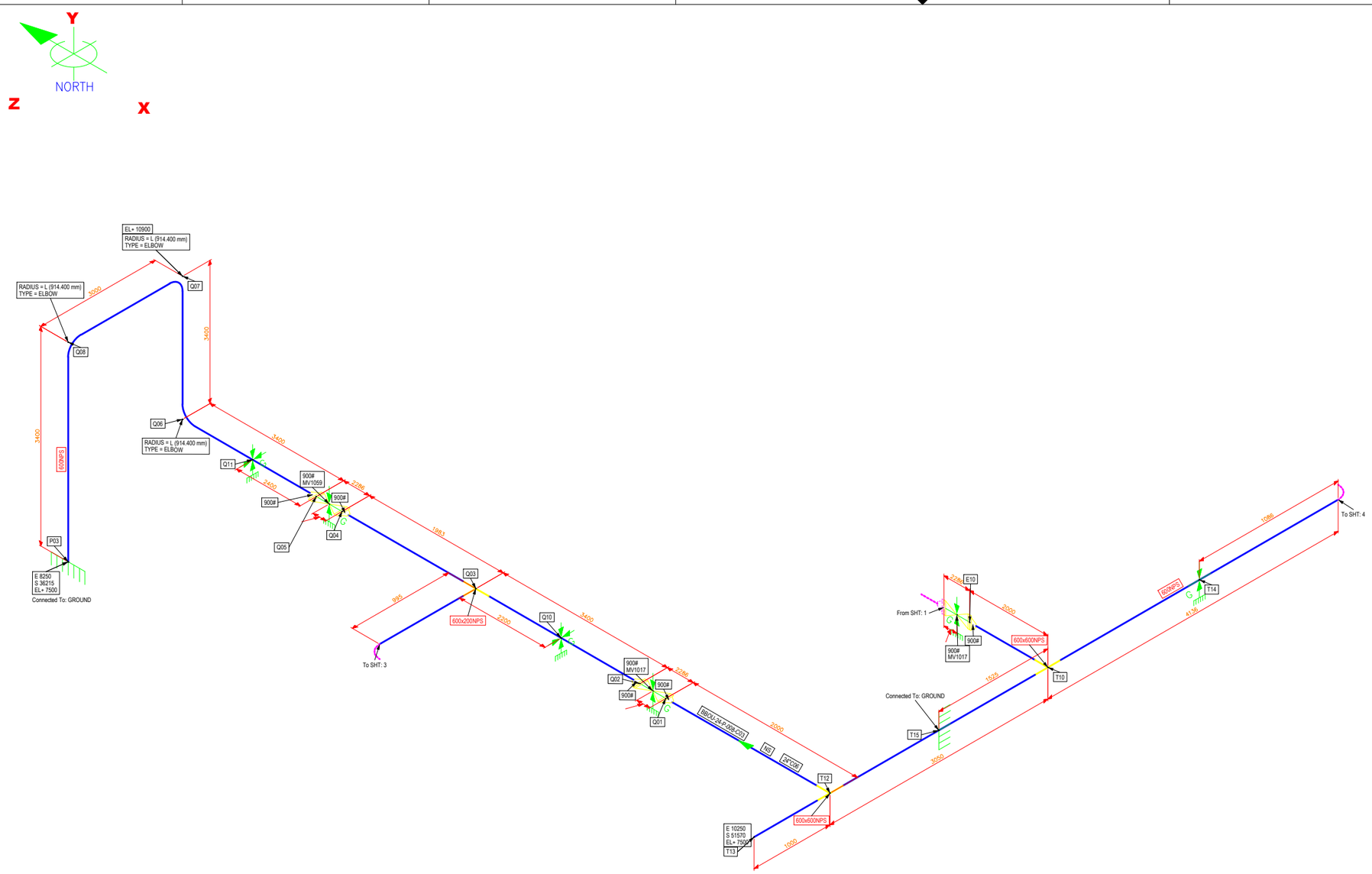
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRG. SIZE D

DRAWING No.

REV.



DRG
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MODEL INFO                    AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code:   ASME B31.3 - 2016
Model Rev: 25          Ambient: 24.0 (deg C)
```

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"C06	API5L-X60	600.00	NS	609.60	38.890	6.000	4.861		0.700
8"C03	API5L-B	200.00	120	219.07	18.263	3.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
E09	600.00	CATALOG	900#	6785.86
Q01	600.00	CATALOG	900#	6785.86
Q04	600.00	CATALOG	900#	6785.86

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
E10	600.00	WELDNECK	900#	1764.00	
Q01	600.00	WELDNECK	900#	1764.00	
Q02	600.00	WELDNECK	900#	1764.00	
Q04	600.00	WELDNECK	900#	1764.00	
Q05	600.00	WELDNECK	900#	1764.00	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
E09 M1	V-stop	Ground	Rigid		2540.00	0.10
Q01 M1	V-stop	Ground	Rigid		2540.00	0.10
Q04 M1	V-stop	Ground	Rigid		2540.00	0.10
Q10 1	Guide-H	Ground		3.00	3.00	0.10
Q10 1	Guide-V	Ground			3.00	0.10
Q11 1	Guide-H	Ground		3.00	3.00	0.10
Q11 1	Guide-V	Ground			3.00	0.10
T14 1	V-stop	Ground	Rigid		2540.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
Q03	AY-Branch	219.07	18.26	API5L-B					
Q03	Q-Header	609.60	38.89	API5L-X60	Welding			1.449	1.598
T12	Q-Branch	609.60	38.89	API5L-X60					
T12	T-Header	609.60	38.89	API5L-X60	Welding			1.449	1.598

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
E09 M1	-2525.37	-161587	-15959.6	-1.01		-6.43	GT1P1(1)
E09 M1	-690.33	7714.08	-20793.9	-0.03		-0.18	Pressure 3(2)
E09 M1	3192.45	-189178	18646.44	0.68		3.90	GT2(1)
E09 M1	-3947.97	-155612	-14440.3	-0.01		-0.08	GT3P3(1)
E09 M1		-159876				-0.03	Gravity(1)
Q01 M1	-9165.58	287.74	11897.61	-0.05		0.06	Pressure 3(1)
Q01 M1	-12599.9	-129375	2936.01	-2.70		0.66	GT1P1(1)
Q01 M1		-113304		0.01		0.03	Gravity(1)
Q04 M1	-19673.0	-28901.4	-4935.82	-9.35		-2.34	Thermal 1(1)
Q04 M1	-19802.9	-204377	-5054.22	-9.83		-2.32	GT1P1(1)
Q04 M1	-37674.8	-3591.49	-3968.55	-0.65		-0.09	Pressure 3(2)
Q04 M1		-173926		0.01		0.20	Gravity(1)
Q10 1				-4.65	-0.07	-0.28	Thermal 1(1)
Q10 1				3.13	0.10	0.22	GT2(1)
Q10 1				-4.85	-0.02	-0.24	GT1P1(1)
Q10 1				0.01	0.05	0.06	Gravity(1)
Q11 1				-12.48	0.83	-2.92	Thermal 1(1)
Q11 1				8.38	-1.19	2.27	GT2(1)
Q11 1				-13.13	0.25	-2.80	GT1P1(1)
Q11 1				0.02	-0.63	0.34	Gravity(1)
T14 1	4139.10	-43689.5	-1398.32	12.95		-4.37	GT1P1(1)
T14 1	4738.43	-53067.1	-2388.46	0.83		-0.42	GT3P3(2)
T14 1	9616.61	-1422.18	-4083.67	0.97		-0.47	Pressure 3(2)
T14 1		-44266.8					Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
P03	-135194	37962.72	22433.89	119720.7	236413.6	217175.4	Thermal 1(1)
P03	85842.66	-66623.8	-14999.8	-106716	-149096	-128244	GT2(1)
P03	-147533	-1007.24	23931.22	100413.9	258739.1	246695.3	GT1P1(1)
T15	-304488	-10265.0	-40058.5	64502.59	549898.4	-7746.90	GT2(1)
T15	461077.3	6211.36	42454.47	-17749.6	-831768	-199.93	Thermal 1(1)
T15	89485.14	-27762.1	19270.67	39144.97	-32059.3	-3499.39	GT3P3(2)
T15	489298.3	-12801.3	42196.15	21528.04	-876275	-4571.97	GT1P1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap — Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

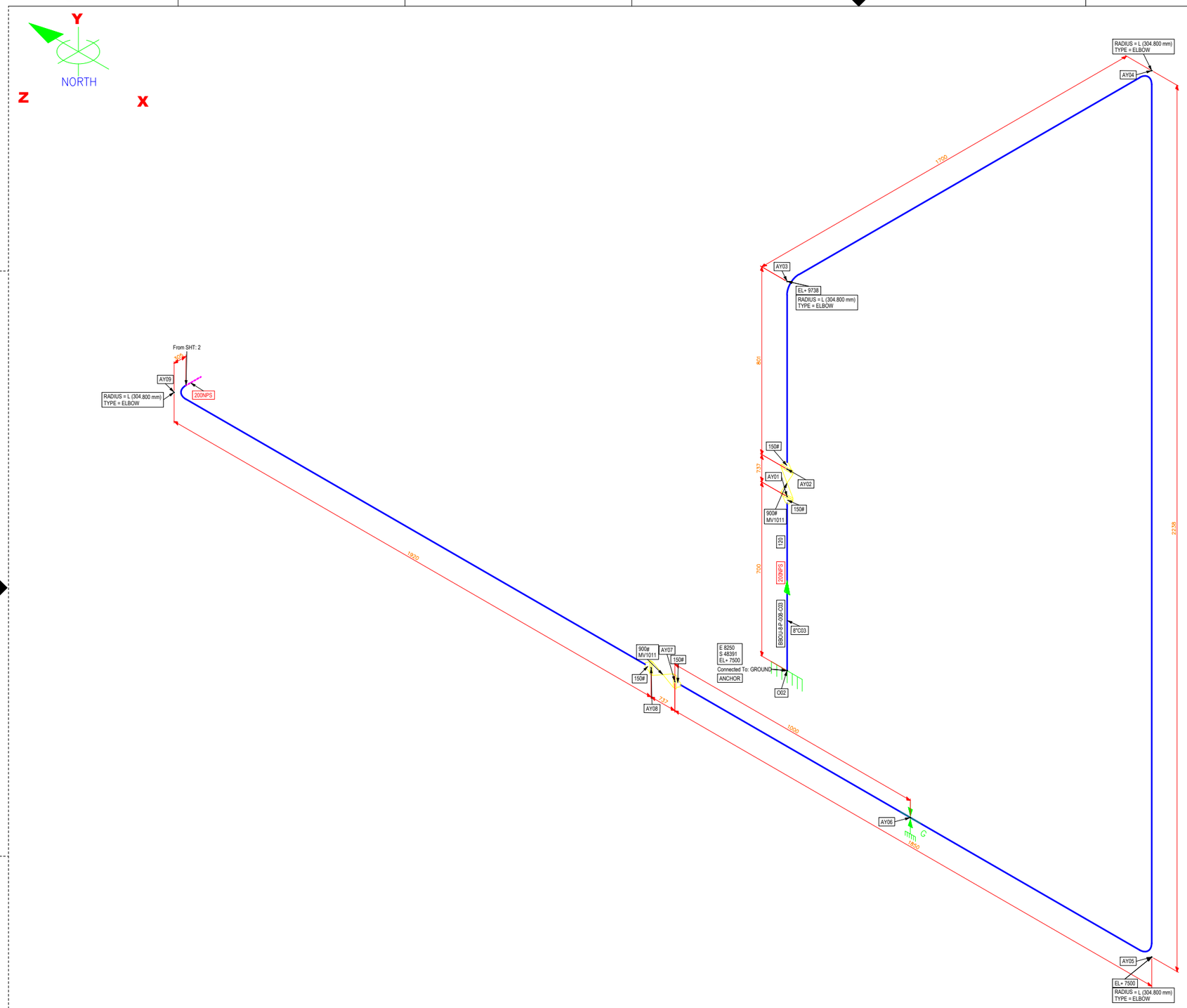
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. D	DATE:
CHECKED BY:	E. N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRAWING No.

DRG. SIZE D

REV.



DRG
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MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
8"C03	API5L-B	200.00	120	219.07	18.263	3.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AY01	200.00	CATALOG	900#	578.79
AY07	200.00	CATALOG	900#	578.79

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AY01	200.00	WELDNECK	150#	39.01	
AY02	200.00	WELDNECK	150#	39.01	
AY07	200.00	WELDNECK	150#	39.01	
AY08	200.00	WELDNECK	150#	39.01	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
AY06 1	V-stop	Ground	Rigid	2540.00		0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AY06 1	-985.77	-10084.0	-212.46	-3.74		-0.95	GTIP1(1)
AY06 1	67.02	-79.00	-447.28			-0.01	Wind 4(1)
AY06 1	-1031.83	-10427.1	-149.89	-0.13		-0.15	GT3P3(2)
AY06 1	-1110.85	-198.14	-170.91	-0.20		-0.03	Pressure 3(2)
AY06 1		-9504.94		0.05		-0.13	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
O02	82.81	-8954.12	422.11	492.16	4.26	-80.82	GW3(1)
O02	130.97	-9321.57	71.51	59.01	-37.78	-185.73	GT3(2)
O02	-2669.52	-9163.88	211.18	-320.25	3445.42	2974.19	GTIP1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

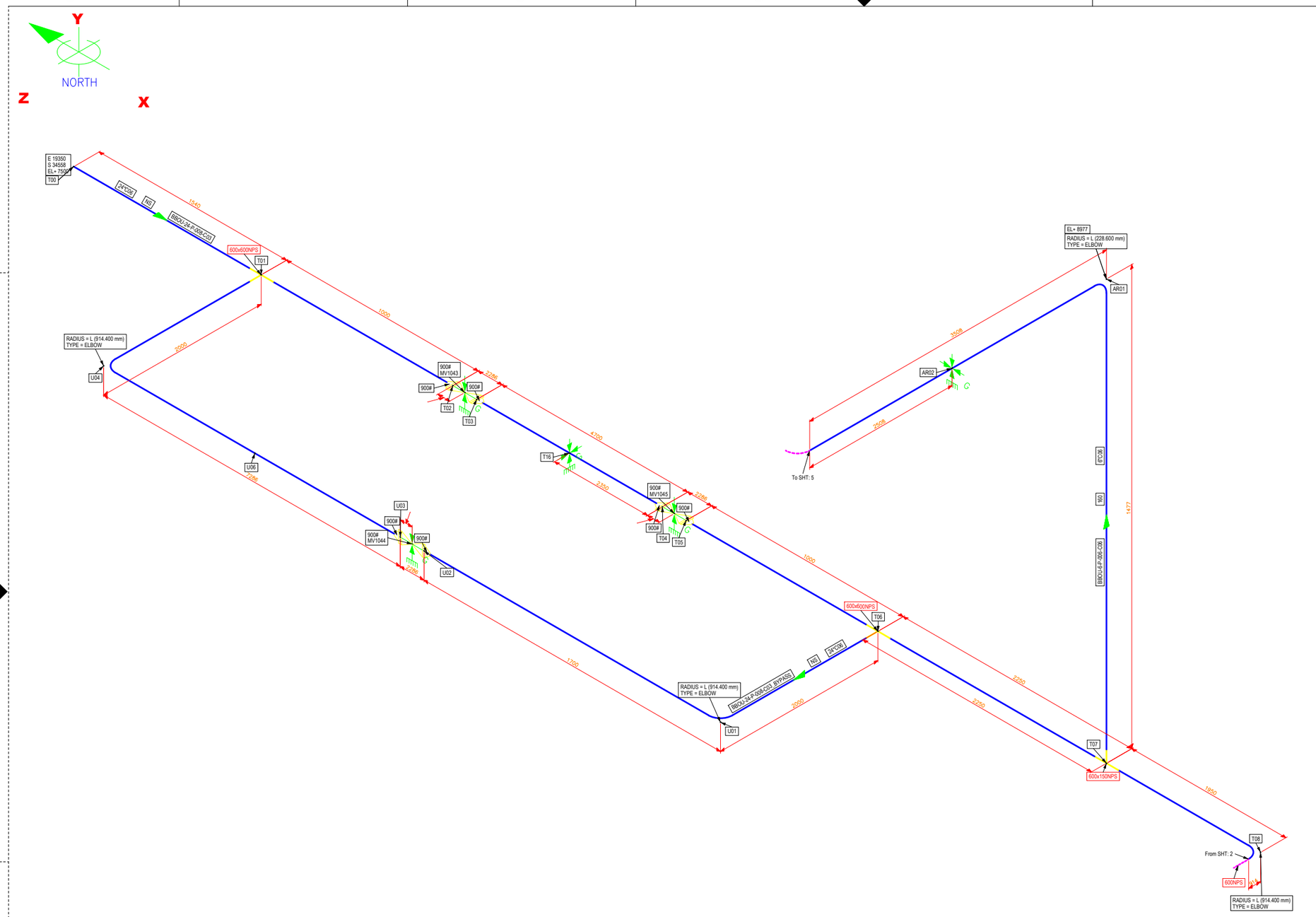
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

REV.



MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"C06	API5L-X60	600.00	NS	609.60	38.890	6.000	4.861		0.700
6"C06	API5L-B	150.00	160	168.28	18.263	6.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
T02	600.00	CATALOG	900#	6785.86
T04	600.00	CATALOG	900#	6785.86
U02	600.00	CATALOG	900#	6785.86

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
T02	600.00	WELDNECK	900#	1764.00	
T03	600.00	WELDNECK	900#	1764.00	
T04	600.00	WELDNECK	900#	1764.00	
T05	600.00	WELDNECK	900#	1764.00	
U02	600.00	WELDNECK	900#	1764.00	
U03	600.00	WELDNECK	900#	1764.00	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
AR02 1	Guide-H	Ground		3.00	3.00	0.10
AR02 1	Guide-V	Ground			3.00	0.10
T02 M1	V-stop	Ground	Rigid		2540.00	0.10
T04 M1	V-stop	Ground	Rigid		2540.00	0.10
T16 1	Guide-H	Ground		3.00	3.00	0.10
T16 1	Guide-V	Ground			3.00	0.10
U02 M1	V-stop	Ground	Rigid		2540.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
T01	U-Branch	609.60	38.89	API5L-X60					
T01	T-Header	609.60	38.89	API5L-X60	Welding			1.449	1.598
T06	U-Branch	609.60	38.89	API5L-X60					
T06	T-Header	609.60	38.89	API5L-X60	Welding			1.449	1.598
T07	AR-Branch	168.27	18.26	API5L-B					
T07	T-Header	609.60	38.89	API5L-X60	Fitting			1.533	1.533

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AR02 1				19.01	0.97	-7.88	Thermal 1(1)
AR02 1				19.75	0.73	-8.42	GT1P1(1)
AR02 1				-0.15	-0.24	-0.06	Gravity(1)
T02 M1	15577.42	-171153	-7090.72	8.48		-3.86	GT1P1(1)
T02 M1	15543.89	-164.55	-7159.33	8.09		-3.72	Thermal 1(1)
T02 M1	18222.03	-182234	195.43	0.46		0.01	GT3P3(2)
T02 M1	36411.82	-127.17	1009.95	0.55		0.01	Pressure 3(2)
T02 M1		-170969					Gravity(1)
T04 M1	10968.91	-127224	-6445.40	14.96		-8.79	GT1P1(1)
T04 M1	-13394.5	-135645	2129.87	-0.15		0.03	GT3(2)
T04 M1	25971.30	285.75	-7134.76	1.01		-0.37	Pressure 3(2)
T04 M1		-128082					Gravity(1)
T16 1				-0.12	0.12		GT3(2)
T16 1				11.72	0.11	-6.42	GT1P1(1)
T16 1					0.12		Gravity(1)
U02 M1	14842.73	-159246	-5769.18	15.38		-5.98	GT1P1(1)
U02 M1	14802.77	-434.98	-5834.83	14.57		-5.74	Thermal 1(1)
U02 M1	16957.17	-170901	-2125.93	0.96		-0.12	GT3P3(2)
U02 M1	33976.90	-209.63	-3245.89	1.07		-0.13	Pressure 3(2)
U02 M1		-158677					Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

DRG
4 OF 17

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap — Line Direction For Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRG. SIZE D

DRAWING No.

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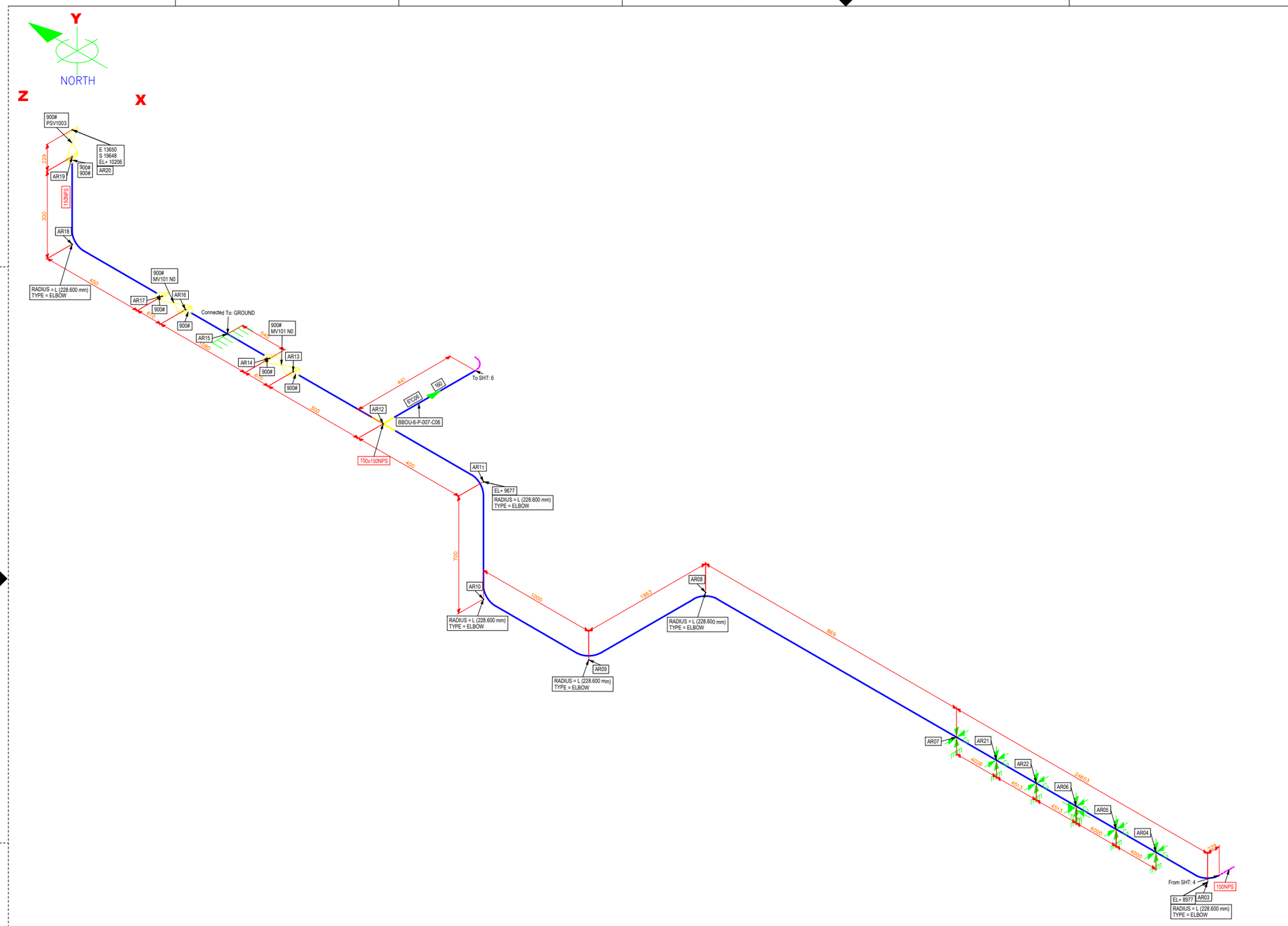

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
6"C06	API5L-B	150.00	160	168.28	18.263	6.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AR13	150.00	CATALOG	900#	359.25
AR16	150.00	CATALOG	900#	359.25
AR19	150.00	CATALOG	900#	359.25

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AR13	150.00	WELDNECK	900#	49.90	
AR14	150.00	WELDNECK	900#	49.90	
AR16	150.00	WELDNECK	900#	49.90	
AR17	150.00	WELDNECK	900#	49.90	
AR19	150.00	WELDNECK	900#	49.90	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
AR04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AR04 1	Guide-V	Ground	Rigid		3.00	0.10
AR05 1	Guide-H	Ground		3.00	3.00	0.10
AR05 1	Guide-V	Ground			3.00	0.10
AR06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AR06 1	Guide-V	Ground	Rigid		3.00	0.10
AR06 2	Linestop	Ground				
AR07 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AR07 1	Guide-V	Ground	Rigid		3.00	0.10
AR21 1	Guide-H	Ground		3.00	3.00	0.10
AR21 1	Guide-V	Ground			3.00	0.10
AR22 1	Guide-H	Ground		3.00	3.00	0.10
AR22 1	Guide-V	Ground			3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
AR12	AT-Branch	168.27	18.26	API5L-B					
AR12	AR-Header	168.27	18.26	API5L-B	Welding			1.064	1.086

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AR04 1	728.47	-5411.66	-1976.88	21.99		-3.00	GT1P1(1)
AR04 1	509.63	-5908.48	-280.93	-0.03		-0.78	GT3P3(2)
AR04 1	1072.18	-28.55	-320.81	0.85		-0.41	Pressure 3(2)
AR04 1		-5380.40		-0.69		-0.34	Gravity(1)
AR05 1				0.37	0.01	2.81	Wind 3(1)
AR05 1				-0.18	-2.62	-1.04	GT3P3(2)
AR05 1				18.29	-2.33	0.93	GT1P1(1)
AR05 1				-0.69	-2.40	-0.81	Gravity(1)
AR06 1	106.24	0.19	1143.46	0.37		4.08	Wind 3(1)
AR06 1	944.45	25.80	59.06	12.81		0.80	Thermal 1(1)
AR06 1	-119.52	0.60	-1218.16	-0.19		-1.92	Wind 4(1)
AR06 1	-590.91	-10379.9	4.66	-0.77		-1.18	GT3(2)
AR06 1	1584.52	3.71	-15.65	0.34			Pressure 3(2)
AR06 1		-9488.75		-0.69		-1.08	Gravity(1)
AR06 2				0.37		4.08	Wind 3(1)
AR06 2				12.81		0.80	Thermal 1(1)
AR06 2				-0.69		-1.08	Gravity(1)
AR07 1	508.12	-8103.59	-1805.44	0.51		-3.00	GT1(1)
AR07 1	508.12	-487.49	-1805.44	1.21		-2.14	Thermal 1(1)
AR07 1	-805.72	-8401.58	-440.28	-0.93		-1.04	GT3P3(2)
AR07 1	-1637.78	-124.90	-695.40	-0.31		-0.14	Pressure 3(2)
AR07 1		-7616.10		-0.70		-0.86	Gravity(1)
AR21 1				0.37	-0.01	4.03	Wind 3(1)
AR21 1				-0.67	-19.25	-0.94	GT3(2)
AR21 1				4.78	0.17	-2.64	Thermal 1(1)
AR21 1				-0.70	-17.59	-0.94	Gravity(1)
AR22 1				0.37	-0.01	5.58	Wind 3(1)
AR22 1				-0.72	-18.59	-1.09	GT3(2)
AR22 1				8.79	0.10	-1.21	Thermal 1(1)
AR22 1				-0.70	-16.98	-1.05	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AR15	223.00	-6528.63	-3231.79	1967.47	3289.12	-7338.10	GT2(1)
AR15	-284.73	-8453.55	-239.21	2314.95	105.91	-5332.87	GT3(2)
AR15	-2866.98	-11128.7	4743.85	1935.11	-5063.40	-859.32	GT1P1(1)
AR15	-2917.45	-11040.7	4622.44	1923.77	-4926.60	-948.50	GT1(1)

DRG
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REV.	DATE	REVISION	BY	CHKD	APP.

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

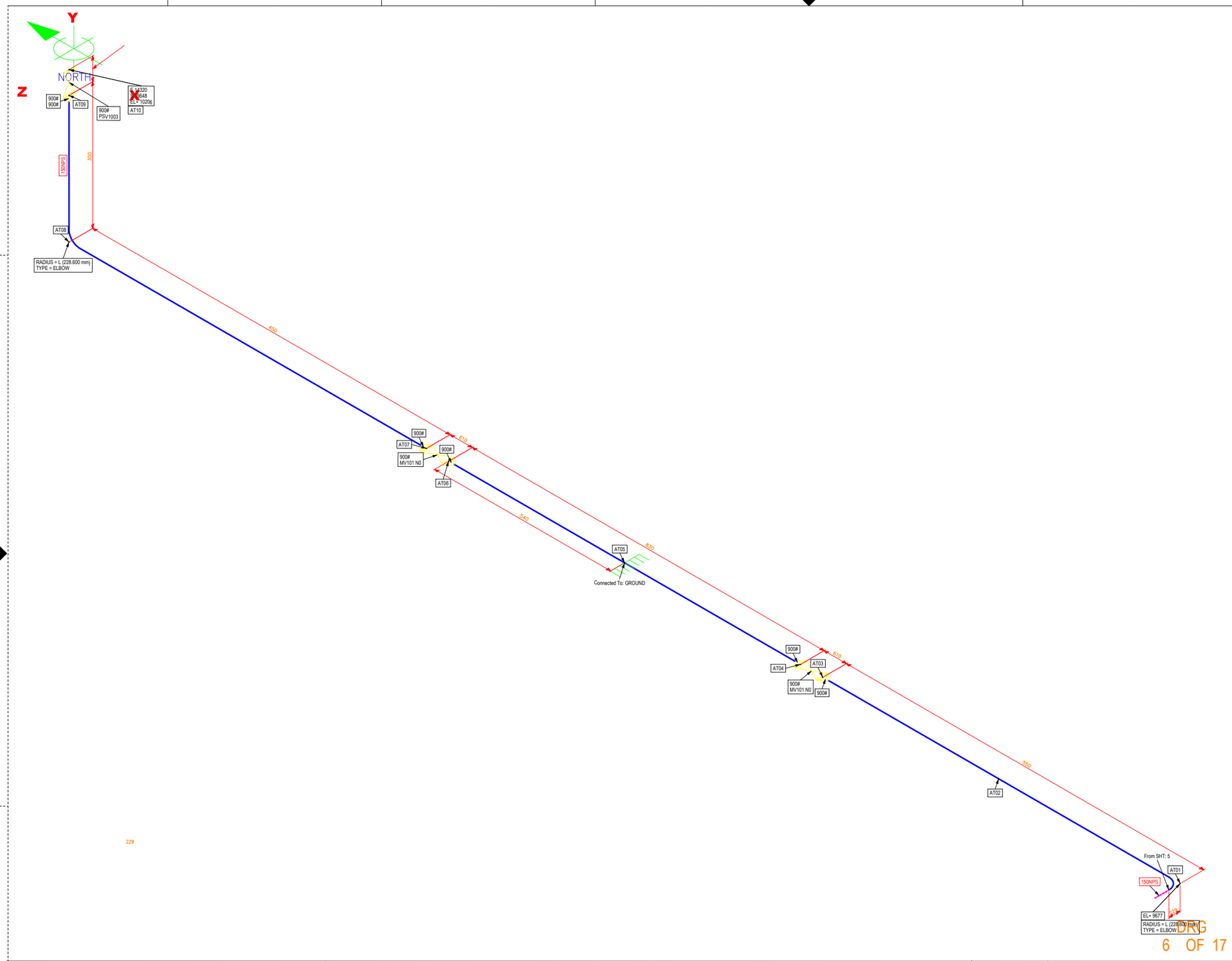
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. O	DATE:
CHECKED BY:	E N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRAWING No.

DRG. SIZE D

REV.



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MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
6"C06	API5L-B	150.00	160	168.28	18.263	6.000	2.283		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AT03	150.00	CATALOG	900#	359.25
AT06	150.00	CATALOG	900#	359.25
AT09	150.00	CATALOG	900#	359.25

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AT03	150.00	WELDNECK	900#	49.90	
AT04	150.00	WELDNECK	900#	49.90	
AT06	150.00	WELDNECK	900#	49.90	
AT07	150.00	WELDNECK	900#	49.90	
AT09	150.00	WELDNECK	900#	49.90	

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AT05	-3310.81	-5107.78	2690.65	1054.66	-1795.37	-6283.70	GT2(1)
AT05	-583.60	-8165.00	-67.84	1443.83	74.00	-3705.46	GT3(2)
AT05	4762.78	-11986.3	-3392.11	1384.59	2725.02	134.62	GT1P1(1)

G-Gap
Line Direction For Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

REV.	DATE	REVISION	BY	CHKD	APP

PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

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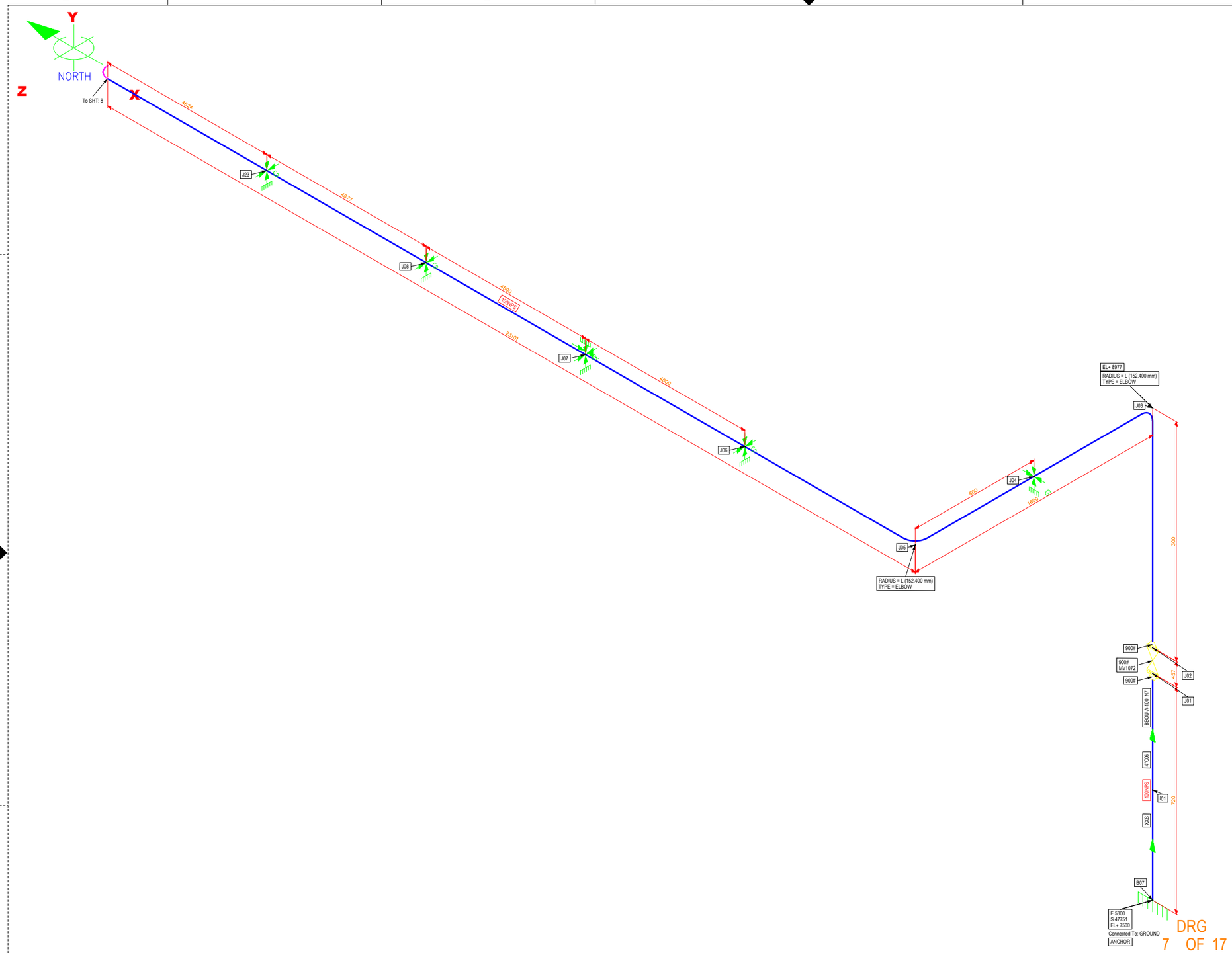

DRG
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MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
J01	100.00	CATALOG	900#	144.70

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
J01	100.00	WELDNECK	900#	23.13	
J02	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
J04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J04 1	Guide-V	Ground	Rigid		3.00	0.10
J06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J06 1	Guide-V	Ground	Rigid		3.00	0.10
J07 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J07 1	Guide-V	Ground	Rigid		3.00	0.10
J07 2	Linestop	Ground	Rigid			
J08 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J08 1	Guide-V	Ground	Rigid		3.00	0.10
J23 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J23 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp, Guide-V=down, Lstop, Incl, Tielink=back, Guide-H=Left
Gap2: Vstp, Guide-V=Up , Lstop, Incl, Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
J04 1	-2271.33	-4902.65	-335.37	-3.00		-1.03	GT2P2(1)
J04 1	12183.36	3474.24	1149.28	3.11	0.31	0.88	Thermal 1(1)
J04 1	-2997.78	-5050.22	-410.74	-3.00		-1.03	GT2(1)
J04 1	12887.82	463.03	1191.86	3.00	0.34	0.80	GT1P1(1)
J04 1		-3074.44		-0.11		-0.07	Gravity(1)
J06 1	-135.85	-6.19	-258.42	-2.37		-3.37	Thermal 2(1)
J06 1	292.97	-2350.42	1179.96	3.60		3.00	GT1P1(1)
J06 1	232.62	-2355.46	37.03	0.08		0.41	GT3P3(2)
J06 1	465.77	-8.28	64.08	0.12		0.02	Pressure 3(2)
J06 1		-2220.72				0.37	Gravity(1)
J07 1		-2.59	-306.58			-3.44	Wind 4(1)
J07 1		1.89	323.35			2.56	Wind 3(1)
J07 1		-1894.08	7.99			0.47	GT3(2)
J07 1	0.01	-12.81	-132.07			-0.02	Thermal 2(1)
J07 1		-1786.54				0.44	Gravity(1)
J07 2	-292.82					-3.44	Wind 4(1)
J07 2	-10750.4					-0.52	GT1P1(1)
J07 2	-76.89					0.44	Gravity(1)
J08 1	-137.38	-1397.17	-23.43	-4.09		-0.33	GT1P1(1)
J08 1	0.24	0.48	-272.21			-3.37	Wind 4(1)
J08 1	-147.44	-1474.81	-3.66	-0.09		0.39	GT3P3(2)
J08 1	-294.86	-0.37	-5.76	-0.14			Pressure 3(2)
J08 1		-1392.14				0.37	Gravity(1)
J23 1	268.46	6.15	-152.24	5.47		-3.10	Thermal 2(1)
J23 1	-329.44	-3095.18	456.47	-8.35		3.00	GT1P1(1)
J23 1	-326.89	-3275.64	20.93	-0.18		0.26	GT3P3(2)
J23 1	-654.21	-0.11	33.47	-0.28		0.02	Pressure 3(2)
J23 1		-3092.66				0.23	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
B07	-1165.56	959.56	643.75	-616.98	-1974.06	1926.24	GT2P2(1)
B07	-1228.88	-4351.46	-2210.37	-726.84	3671.45	1904.20	GT1P1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

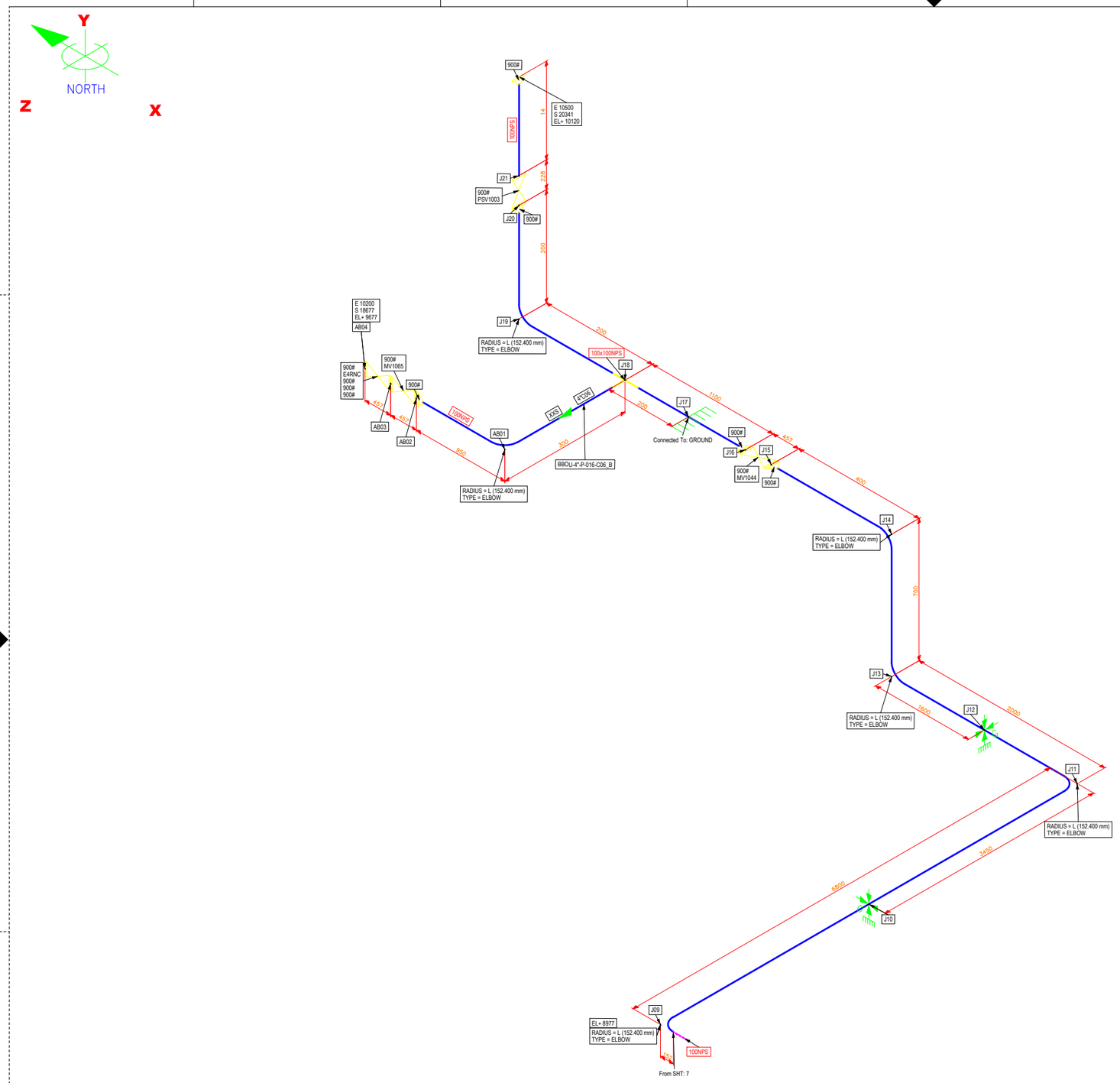
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. O	DATE:
CHECKED BY:	E N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRAWING No.

DRG. SIZE D

REV.



DRG
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MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AB02	100.00	CATALOG	900#	144.70
AB03	100.00	CATALOG	900#	192.78
J15	100.00	CATALOG	900#	144.70
J20	100.00	CATALOG	900#	144.70

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AB02	100.00	WELDNECK	900#	23.13	
J15	100.00	WELDNECK	900#	23.13	
J16	100.00	WELDNECK	900#	23.13	
J20	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
J10 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J10 1	Guide-V	Ground	Rigid		3.00	0.10
J12 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
J12 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
J18	AB-Branch	114.30	17.12	API5L-B					
J18	J-Header	114.30	17.12	API5L-B	Other			1.000	1.000

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
J10 1	-537.21	-3590.82	197.48	-3.00		1.82	GT1(1)
J10 1	-135.83	-1.63	-271.68	-0.01		-0.02	Wind 4(1)
J10 1	-37.35	-3742.24	28.88	-0.05		0.18	GT3P3(2)
J10 1	-574.83	-3592.15	199.87	-3.00		1.80	GT1P1(1)
J10 1		-3532.96		-0.05		0.17	Gravity(1)
J12 1	50.47	-60.29	-24.83	3.10		-1.53	Thermal 1(1)
J12 1	50.47	-562.37	-24.83	2.99		-1.35	GT1(1)
J12 1	73.32	-15.92	-72.31	0.09		-0.09	Pressure 3(2)
J12 1		-502.08		-0.11		0.17	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
J17	-1526.53	-4773.97	172.95	228.19	-1469.45	-1888.65	GT2(1)
J17	29.96	-6160.67	306.24	201.07	-95.82	-1151.60	GW3(1)
J17	1054.69	-6972.96	-301.22	363.66	2565.84	367.72	GT1P1(1)
J17	-1563.06	1395.58	170.95	28.44	-1448.69	-736.81	Thermal 2(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

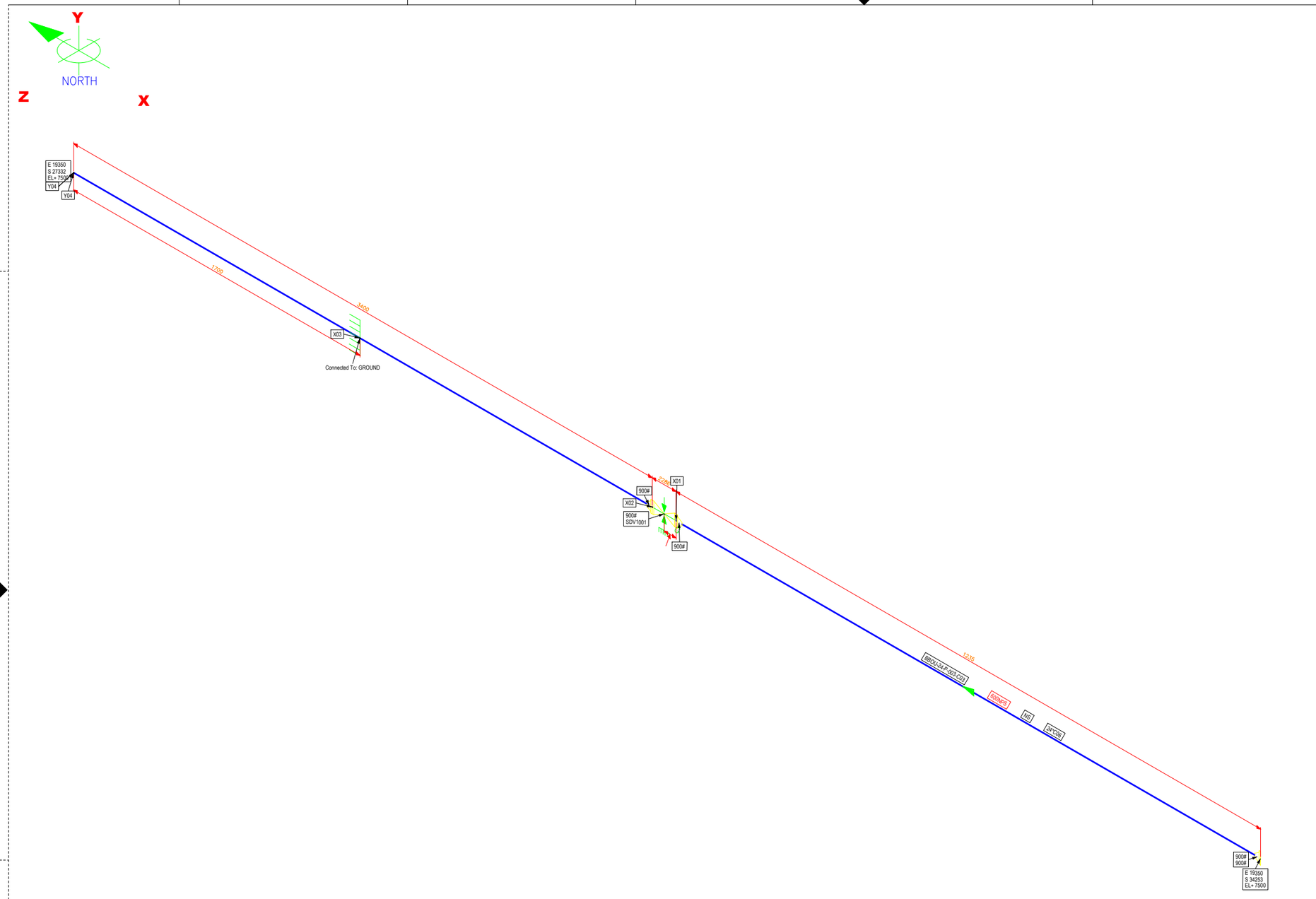
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	T. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

REV.



MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24"C06	API5L-X60	600.00	NS	609.60	38.890	6.000	4.861		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
X01	600.00	CATALOG	900#	6785.86

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
X01	600.00	WELDNECK	900#	1764.00	882.00
X02	600.00	WELDNECK	900#	1764.00	882.00

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
X01 M1	V-stop	Ground	Rigid	2540.00		0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
X01 M1	22841.64	-235475	-5722.40	2.61		-0.65	GT1P1(1)
X01 M1	-22681.0	-245990	2757.73	-0.02			GT3(2)
X01 M1	47246.05	134.40	-3753.94	0.16		-0.01	Pressure 3(2)
X01 M1		-235733					Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
X03	68288.20	-37685.2	-849.27	15510.06	-96.77	159404.8	GT3(2)
X03	-71776.4	-37757.0	2706.35	15570.72	21780.58	159360.0	GT3P3(2)
X03	-195354	-33524.1	12820.40	12921.13	267885.7	145907.1	GT1P1(1)

DRG
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REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

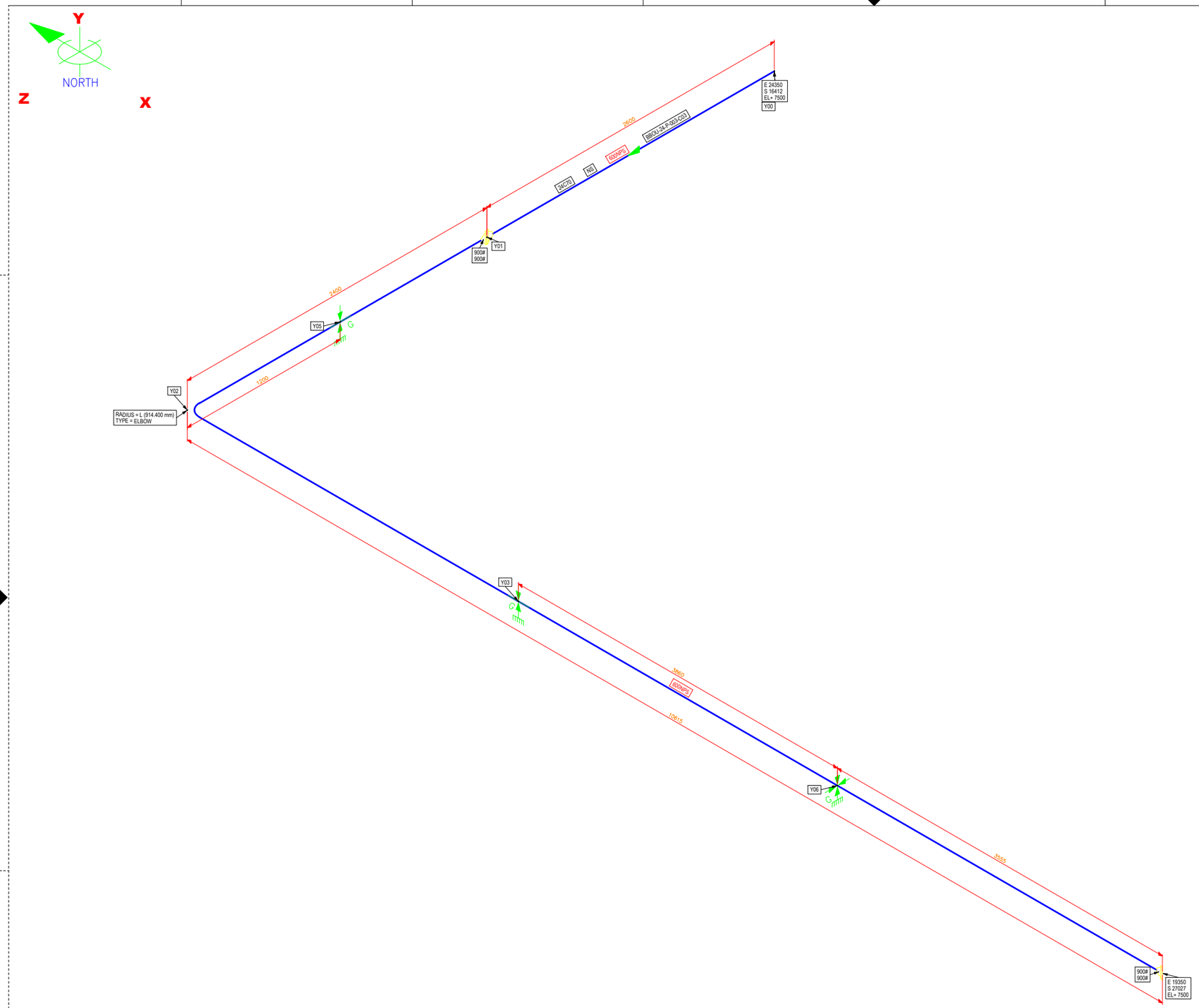
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I.O	DATE:	
CHECKED BY:	E.N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

REV.



DRG
10 OF 17

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24C70	A790-S31803	600.00	NS	609.60	24.200		3.025		0.700

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
Y01	600.00	WELDNECK	900#	1764.00	882.00
Y01	600.00	WELDNECK	900#	1764.00	882.00

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
Y03 1	V-stop	Ground	Rigid		2540.00	0.10
Y05 1	V-stop	Ground	Rigid		2540.00	0.10
Y06 1	Guide-H	Ground		3.00	3.00	0.10
Y06 1	Guide-V	Ground			3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
Y03 1	-2926.81	-35845.4	2069.47	-10.01		7.07	GT1P1(1)
Y03 1		-44870.2					Gravity(2)
Y03 1	-7675.76		4590.65	-1.60		1.11	Pressure 3(2)
Y03 1		-35845.4					Gravity(1)
Y05 1	-4569.93	-51496.8	2373.82	-10.76		5.59	GT1P1(1)
Y05 1		-55132.6					Gravity(2)
Y05 1	-9618.58		5390.17	-1.73		0.95	Pressure 3(2)
Y05 1		-51496.8					Gravity(1)
Y06 1					-0.61		Gravity(2)
Y06 1				-5.81	-0.52	3.41	GT1P1(1)
Y06 1					-0.52		Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

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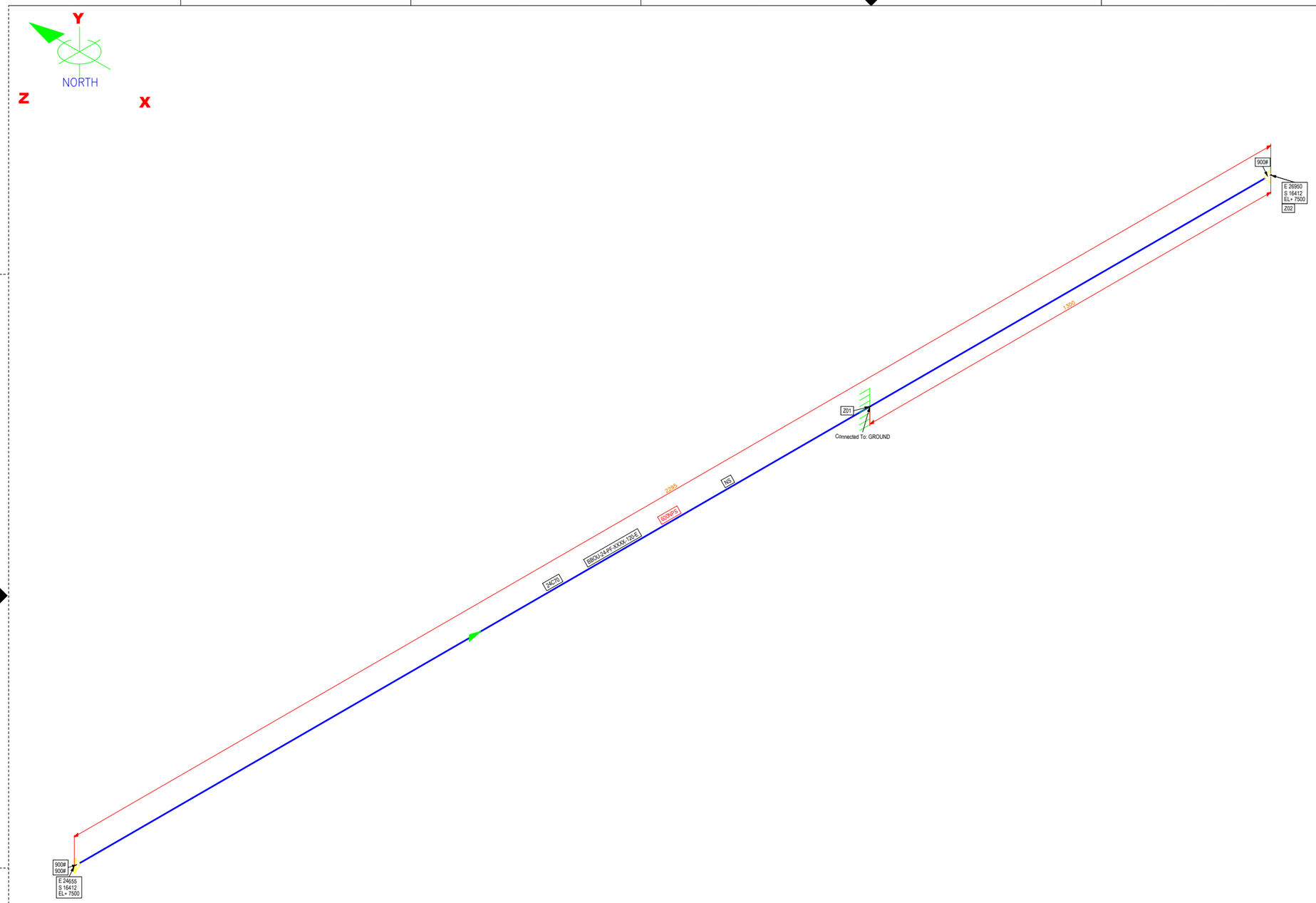


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MODEL INFO                    AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code:   ASME B31.3 - 2016
Model Rev: 25          Ambient: 24.0 (deg C )
```

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
24C70	A790-S31803	600.00	NS	609.60	24.200		3.025		0.700

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
Z02	600.00	WELDNECK	900#	1764.00	

```
 Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
 Gap2: Vstp,Guide-V=Up  , Lstop,Incl,Tielink=frwd, Guide-H=Right
 Gap 3 & 4: Guides (Guide-H) only

 For Rotational Supports: Loads are moments (ft-lb) and
 movements are rotations (deg)

 Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
         Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
 Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
 Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
 Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)
```

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
Z01	61585.83	-85788.6	19339.97	43281.39	291329.3	-905.93	GT2(1)
Z01		-91168.3		45914.17	0.01	-871.58	Gravity(2)
Z01	-99063.0	-85788.6	-28588.5	43281.39	-467532	-905.92	GT1P1(1)

DRG
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REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I.O	DATE:	
CHECKED BY:	E.N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRG. SIZE D

DRAWING No.

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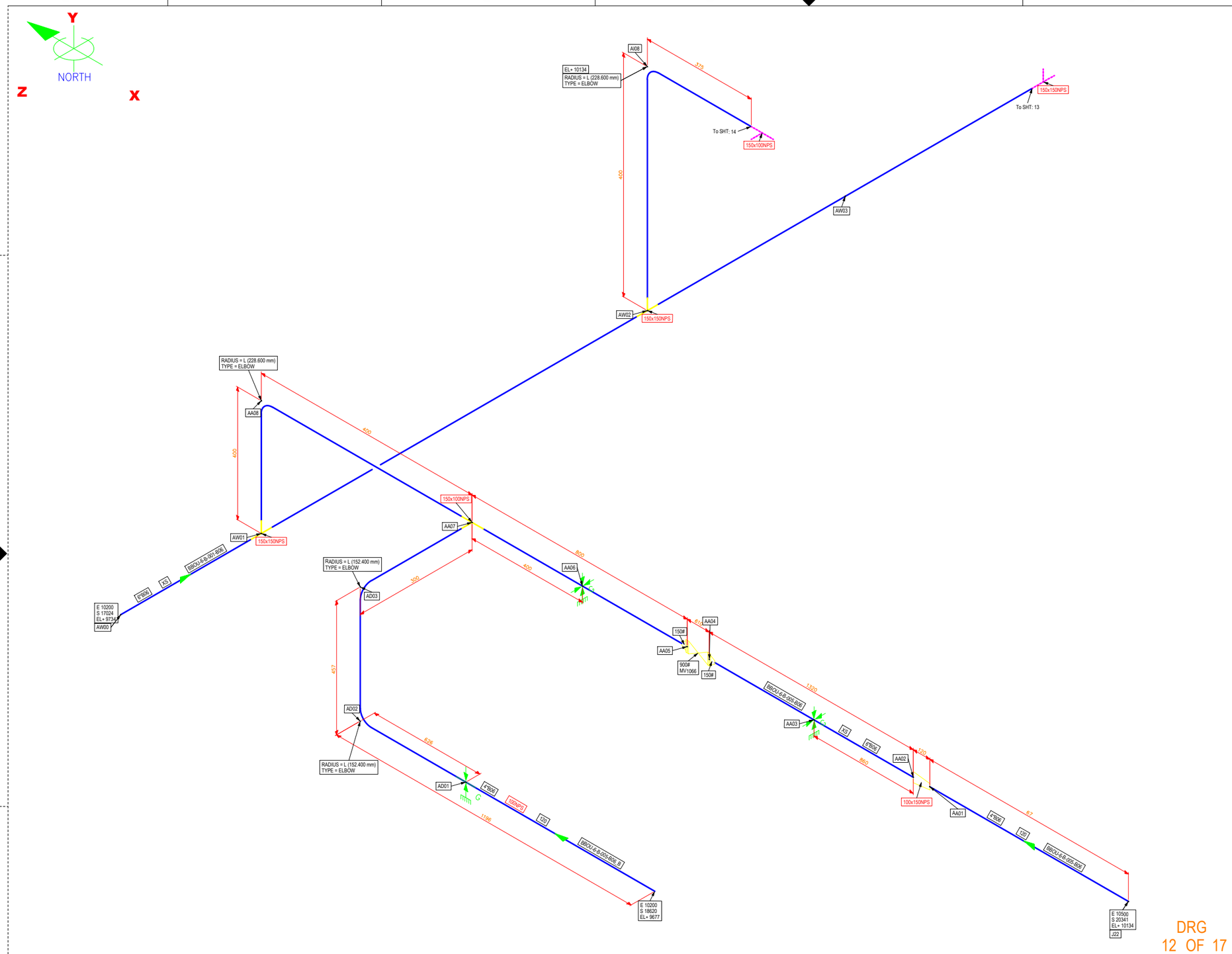

DRG
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MODEL INFO                    AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code:   ASME B31.3 - 2016
Model Rev: 25          Ambient: 24.0 (deg C )
```

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4" B06	API5L-B	100.00	120	114.30	11.125	6.000	1.391		0.700
6" B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AA04	150.00	CATALOG	900#	359.25

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AA04	150.00	WELDNECK	150#	11.34	
AA05	150.00	WELDNECK	150#	11.34	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
AA03 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AA03 1	Guide-V	Ground	Rigid		3.00	0.10
AA06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AA06 1	Guide-V	Ground	Rigid		3.00	0.10
AD01 1	V-stop	Ground	Rigid		2540.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
AA07	AD-Branch	114.30	11.13	API5L-B					
AA07	AA-Header	168.27	10.97	API5L-B	Welding			1.430	1.574

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AA03 1	-223.10	-2353.31	74.90	-1.05		0.35	GT1P1(1)
AA03 1	508.37	-5430.48	-190.94	0.92		-0.35	GT2(1)
AA03 1	508.37	-2217.14	-190.94	0.93		-0.35	Thermal 2(1)
AA03 1		-3213.35		-0.01		0.01	Gravity(1)
AA06 1	296.44	-3310.66	-147.40	1.80		-0.92	GT2(1)
AA06 1		2099.67		-1.63	0.15	0.65	Thermal 1(1)
AA06 1	296.44	-1210.99	-147.40	1.81		-0.90	Thermal 2(1)
AA06 1	-1.56	-77.88	-217.56			-0.03	Wind 4(1)
AA06 1	296.83	-3312.33	-146.99	1.77		-0.90	GT2P2(1)
AA06 1	-309.12	4.66	91.63	-0.04		0.01	Pressure 3(2)
AA06 1		-2099.67		-0.01		-0.01	Gravity(1)
AD01 1	120.98	1587.79	-82.49	1.55		-1.06	Thermal 2(1)
AD01 1	33.24	-27.26	217.96			0.02	Wind 3(1)
AD01 1	-419.87	-4611.42	190.72	-2.01		0.92	GT1P1(1)
AD01 1	-428.92	-38.45	92.22	-0.04		0.01	Pressure 3(2)
AD01 1		-3052.05					Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. O	DATE:
CHECKED BY:	E. N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRG. SIZE D

DRAWING No.

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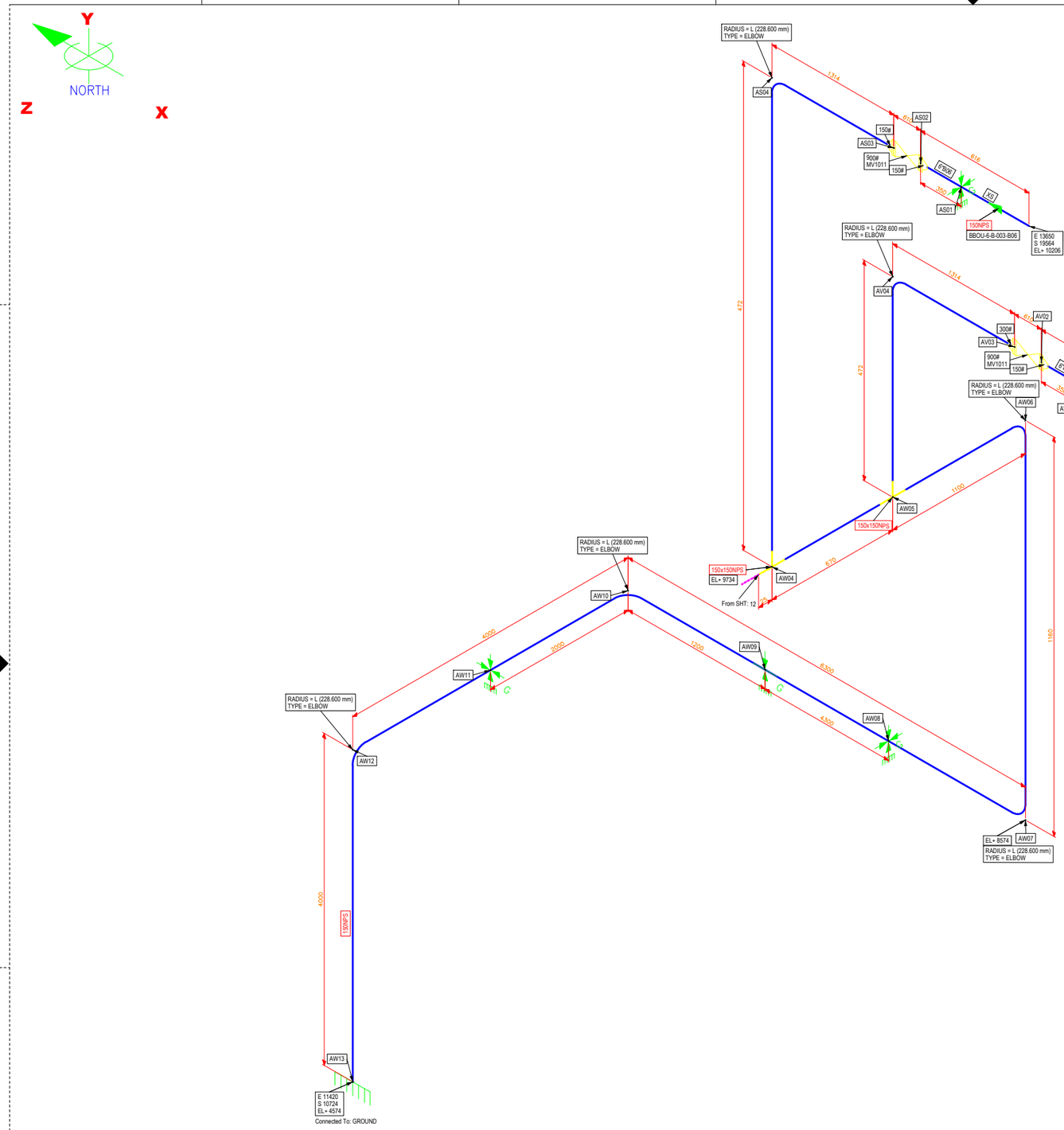


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MODEL INFO                    AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code:   ASME B31.3 - 2016
Model Rev: 25          Ambient: 24.0 (deg C)
```

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
6"B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AS02	150.00	CATALOG	900#	359.25
AV02	150.00	CATALOG	900#	359.25

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AS02	150.00	WELDNECK	150#	11.34	
AS03	150.00	WELDNECK	150#	11.34	
AV02	150.00	WELDNECK	150#	11.34	
AV03	150.00	WELDNECK	300#	18.60	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
AS01 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AS01 1	Guide-V	Ground	Rigid		3.00	0.10
AV01 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AV01 1	Guide-V	Ground	Rigid		3.00	0.10
AW08 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AW08 1	Guide-V	Ground	Rigid		3.00	0.10
AW09 1	V-stop	Ground	Rigid		2540.00	0.10
AW11 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AW11 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AS01 1	-712.18	-7155.29	-69.49	-1.68		-0.18	GT1P1(1)
AS01 1	6.82	3.56	318.90			0.01	Wind 3(1)
AS01 1	1408.12	-14182.6	169.26	1.24		0.13	GT2(1)
AS01 1	-1544.26	588.10	-37.60	-0.03			Pressure 3(2)
AS01 1		-11270.4		0.02		-0.02	Gravity(1)
AV01 1	-682.10	-6853.94	-67.82	-1.68		-0.15	GT1P1(1)
AV01 1	-14.45	-1.36	312.36			0.01	Wind 3(1)
AV01 1	1512.25	-15309.1	238.31	1.20		0.21	GT2(1)
AV01 1	1512.25	-4071.99	238.31	1.18		0.19	Thermal 2(1)
AV01 1		-11237.1		0.02		0.02	Gravity(1)
AW08 1	181.86	696.50	212.56	2.25		2.63	Thermal 2(1)
AW08 1	16.10	-3.57	-348.13	0.01		-0.13	Wind 4(1)
AW08 1	-359.85	-4575.25	-282.18	-2.51		-2.15	GT1P1(1)
AW08 1		-3493.90		-0.14		-0.30	Gravity(1)
AW09 1	18.42	1461.09	13.64	4.82		3.57	Thermal 2(1)
AW09 1	-3.09	-24.77	171.06	-0.01		0.41	Wind 3(1)
AW09 1	-198.02	-2279.75	-112.95	-4.26		-2.43	GT1(1)
AW09 1	-335.75	-80.34	-90.66	-0.06		-0.02	Pressure 3(2)
AW09 1		-1690.31		-0.14		-0.08	Gravity(1)
AW11 1	-369.69	9.71	-35.71	-3.00	0.32	-1.19	GT1P1(1)
AW11 1	815.12	-4649.41	277.17	3.00		1.79	GT2(1)
AW11 1	815.12	-3203.63	277.17	3.09		1.82	Thermal 2(1)
AW11 1		-1445.78		-0.09		-0.03	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AW13	-201.32	-3236.58	289.99	237.63	486.95	783.07	GT1(1)
AW13	75.99	2061.78	-525.34	-447.01	-588.01	-452.48	Thermal 2(1)
AW13	-3.87	-90.53	348.61	510.46	-0.16	15.73	Wind 3(1)
AW13	79.07	-178.41	-528.33	-464.37	-580.61	-433.09	GT2(1)
AW13	-187.51	-3245.52	302.66	265.67	503.48	734.49	GT1P1(1)
AW13	408.07	-2305.08	36.39	78.68	107.46	-757.75	GW1(1)

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REV.	DATE	REVISION	BY	CHKD	APP

G-Gap — Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

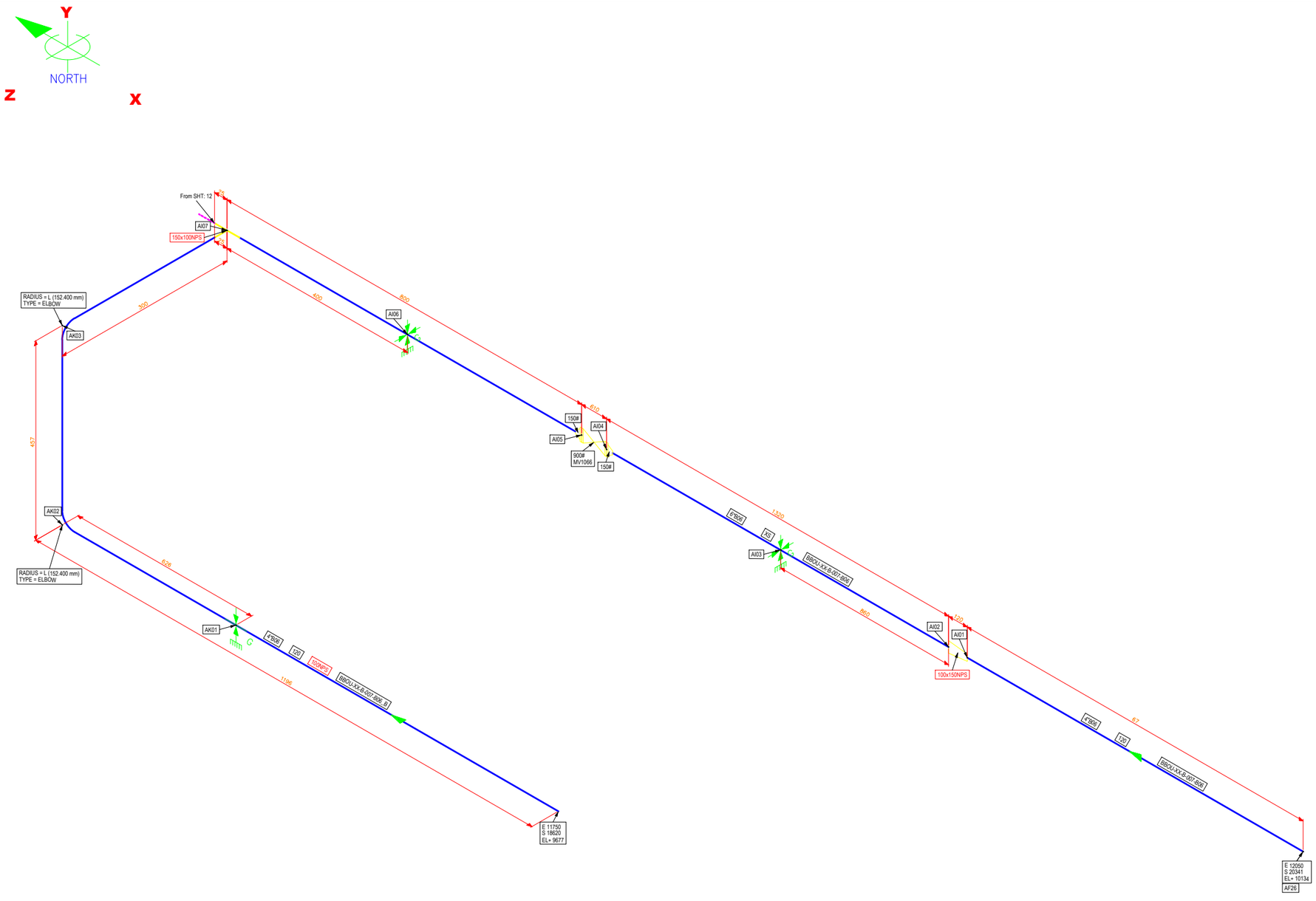
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I.O	DATE:
CHECKED BY:	E N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRAWING No.

DRG. SIZE D

REV.



DRG
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MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"B06	API5L-B	100.00	120	114.30	11.125	6.000	1.391		0.700
6"B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AI04	150.00	CATALOG	900#	359.25

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AI04	150.00	WELDNECK	150#	11.34	
AI05	150.00	WELDNECK	150#	11.34	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
AI03 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AI03 1	Guide-V	Ground	Rigid		3.00	0.10
AI06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AI06 1	Guide-V	Ground	Rigid		3.00	0.10
AK01 1	V-stop	Ground	Rigid		2540.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
AI07	AK-Branch	114.30	11.13	API5L-B					
AI07	AI-Header	168.27	10.97	API5L-B	Welding			1.430	1.574

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AI03 1	-338.49	-1009.99	23.72	-1.07		0.08	Thermal 1(1)
AI03 1	-336.11	-3369.30	23.29	-1.12		0.07	GT1P1(1)
AI03 1	8.95	13.14	-81.20			-0.01	Wind 4(1)
AI03 1	412.99	-4133.10	-16.36	0.95		-0.05	GT2(1)
AI03 1	412.99	-1749.91	-16.36	0.98		-0.04	Thermal 2(1)
AI03 1		-2383.19		-0.03		-0.01	Gravity(1)
AI06 1	492.49	-4942.94	-42.17	1.83		-0.21	GT2(1)
AI06 1	492.49	236.16	-42.17	1.86		-0.16	Thermal 2(1)
AI06 1	-20.28	-21.98	460.08			0.02	Wind 3(1)
AI06 1	557.01	-5839.42	11.55	-0.01		-0.05	GT3(2)
AI06 1	-739.39	48.50	-5.95	-0.03			Pressure 3(2)
AI06 1		-5179.10		-0.03		-0.05	Gravity(1)
AK01 1	36.49	-13.69	207.78			0.02	Wind 3(1)
AK01 1	-589.87	-6048.52	133.82	-2.02		0.48	GT1P1(1)
AK01 1		-3263.91		0.01		0.02	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

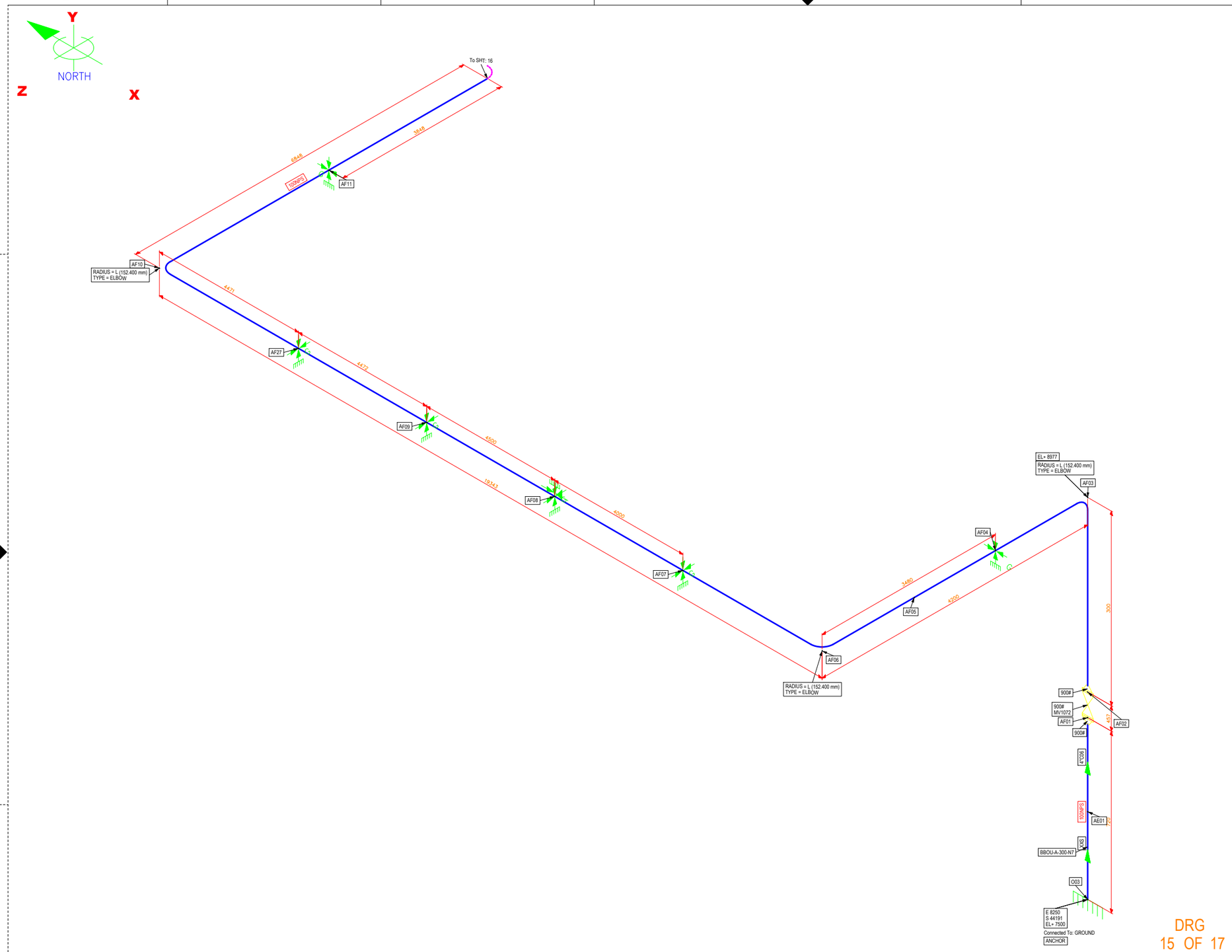
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I.O	DATE:
CHECKED BY:	E.N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRG. SIZE D

DRAWING No.

REV.



DRG
15 OF 17

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AF01	100.00	CATALOG	900#	144.70

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AF01	100.00	WELDNECK	900#	23.13	
AF02	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
AF04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF04 1	Guide-V	Ground	Rigid		3.00	0.10
AF07 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF07 1	Guide-V	Ground	Rigid		3.00	0.10
AF08 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF08 1	Guide-V	Ground	Rigid		3.00	0.10
AF08 2	Linestop	Ground	Rigid			
AF09 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF09 1	Guide-V	Ground	Rigid		3.00	0.10
AF11 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF11 1	Guide-V	Ground	Rigid		3.00	0.10
AF27 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF27 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AF04 1				1.40	0.71	0.98	GT1P1(1)
AF04 1	-344.94	-7151.80	-626.51	-0.40		-1.09	GT2(1)
AF04 1	-344.94	-4319.92	-626.51	-0.57		-1.03	Thermal 2(1)
AF04 1		-2831.88		0.16		-0.06	Gravity(1)
AF07 1	234.78	-245.70	600.87	3.56		3.15	Thermal 1(1)
AF07 1	242.28	-2549.02	657.36	3.64		3.00	GT1P1(1)
AF07 1	234.78	-2554.65	600.87	3.56		3.00	GT1(1)
AF07 1	481.06	-17.76	86.02	0.12		0.02	Pressure 3(2)
AF07 1		-2308.95				-0.15	Gravity(1)
AF08 1		0.68	164.86			3.15	Wind 3(1)
AF08 1		-0.59	-166.55			-2.74	Wind 4(1)
AF08 1		-1813.04	-166.54			-0.47	GT2(1)
AF08 1		-1648.42				-0.26	Gravity(1)
AF08 2	-4.60					3.15	Wind 3(1)
AF08 2	569.42					-0.24	GT1P1(1)
AF08 2	5.28					-0.26	Gravity(1)
AF09 1	-0.32	-0.10	313.12			3.29	Wind 3(1)
AF09 1	-148.24	-1484.87	-8.92	-4.09		-0.54	GT1P1(1)
AF09 1	0.31	0.08	-327.35			-2.71	Wind 4(1)
AF09 1	-155.42	-1554.91	-4.73	-0.09		-0.31	GT3P3(2)
AF09 1	-310.80	-0.84	-7.70	-0.14			Pressure 3(2)
AF09 1		-1467.34				-0.29	Gravity(1)
AF11 1	257.06	-3238.04	-197.04	2.59		-2.16	GT2(1)
AF11 1	-726.01	-3280.61	230.81	-3.00		1.89	GT1(1)
AF11 1	20.36	-3448.74	-59.75	-0.17		-0.06	GT3(2)
AF11 1	-761.61	-3281.47	232.05	-3.00		1.86	GT1P1(1)
AF11 1		-3254.21		-0.16		-0.05	Gravity(1)
AF27 1	-299.89	6.74	474.10	-7.97		3.22	Thermal 1(1)
AF27 1	-303.05	-2836.75	495.06	-8.16		3.00	GT1P1(1)
AF27 1	-300.16	-3011.17	24.00	-0.17		-0.22	GT3P3(2)
AF27 1	-600.92	-0.02	38.60	-0.27		0.02	Pressure 3(2)
AF27 1		-2843.38				-0.22	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
O03	-36.23	3263.08	912.25	-653.43	-525.38	-107.29	GT2(1)
O03	-28.98	4030.25	912.22	-556.01	-501.63	162.09	Thermal 2(1)
O03	544.21	-3445.57	-727.68	219.54	898.81	-1097.77	GT1P1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

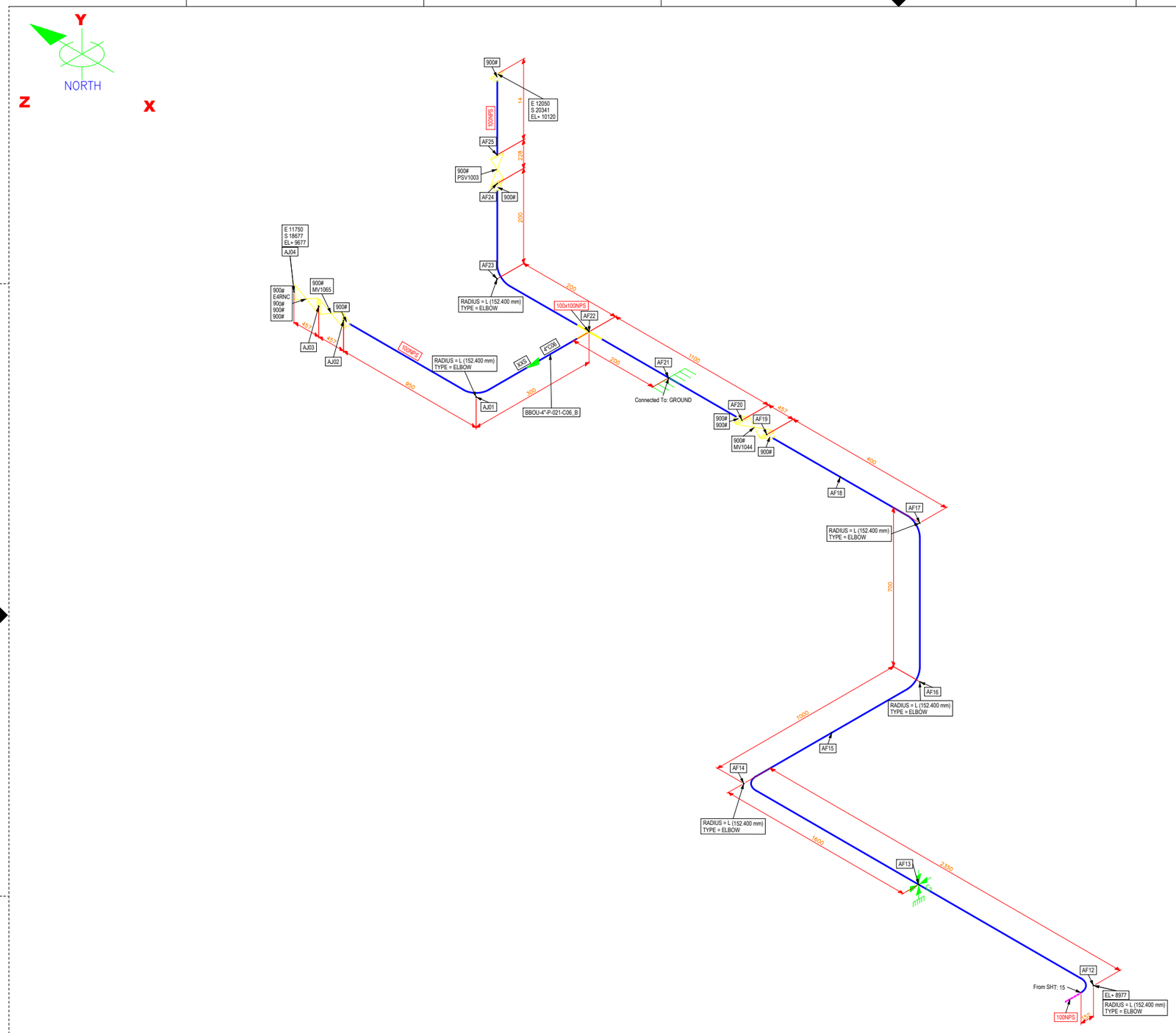
PROJECT NO:	AKABEL	DATE:	
ANALYZED BY:	I. O	DATE:	
CHECKED BY:	E. N	DATE:	
APPROVED BY:		DATE:	
AREA:			
LINE NO.:			
CLASS:			
FLOW DIAGRAM:			

Bentley

DRAWING No.

DRG. SIZE D

REV.



DRG
16 OF 17

MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
4"C06	API5L-B	100.00	XXS	114.30	17.120	6.000	2.140		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AF19	100.00	CATALOG	900#	144.70
AF24	100.00	CATALOG	900#	144.70
AJ02	100.00	CATALOG	900#	144.70
AJ03	100.00	CATALOG	900#	192.78

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AF19	100.00	WELDNECK	900#	23.13	11.56
AF20	100.00	WELDNECK	900#	23.13	
AF24	100.00	WELDNECK	900#	23.13	
AJ02	100.00	WELDNECK	900#	23.13	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn. to	Stiff.	Gap 1	Gap 2	Friction
AF13 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AF13 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

TEE DATA

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	Crotch Radius (mm)	Crotch Thick (mm)	Sif in	Sif out
AF22	AJ-Branch	114.30	17.12	API5L-B					
AF22	AF-Header	114.30	17.12	API5L-B	Welding			1.000	1.000

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AF13 1	128.92	-123.47	-48.71	3.82		-1.44	Thermal 1(1)
AF13 1	128.44	-1379.58	-50.25	3.55		-1.55	GT1P1(1)
AF13 1	139.83	-29.22	-162.50	0.03		-0.03	Pressure 3(2)
AF13 1		-1254.89		-0.38		-0.02	Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AF21	1225.84	-5994.62	467.56	260.23	-706.12	-2662.76	GT2(1)
AF21	1125.82	-5991.50	449.98	262.72	-695.72	-2644.31	GT2P2(1)
AF21	212.97	-6827.46	-11.49	217.62	115.43	-1929.39	GT3P3(2)
AF21	-1707.64	-5707.02	-656.04	184.10	2437.90	-439.91	GT1P1(1)

REV.	DATE	REVISION	BY	CHKD	APP

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

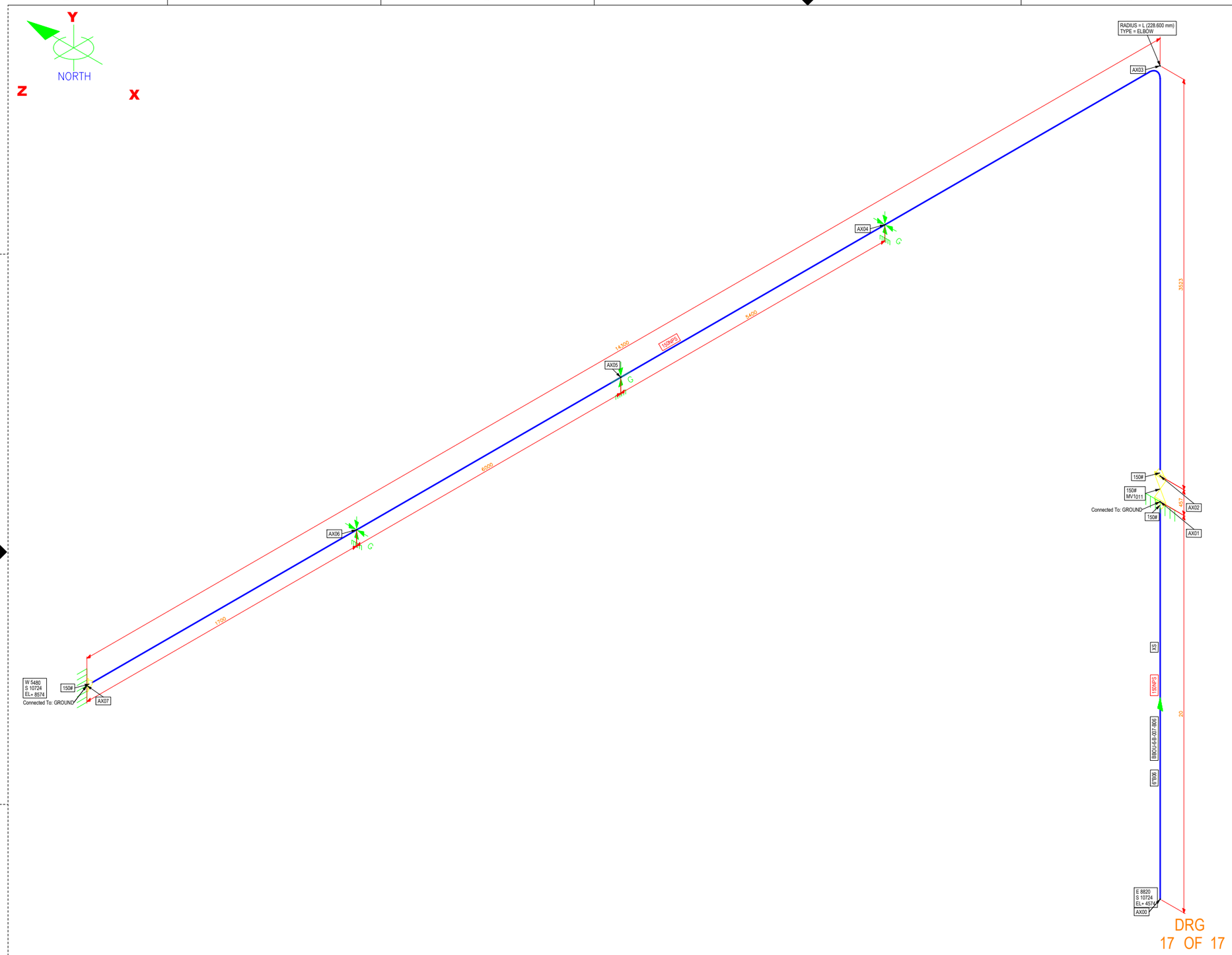
PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I. O	DATE:
CHECKED BY:	E. N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRAWING No.

DRG. SIZE D

REV.



MODEL INFO AutoPIPE Ver: 12.06.00.48
System: BBOU._ASME B31.3
Piping Code: ASME B31.3 - 2016
Model Rev: 25 Ambient: 24.0 (deg C)

PIPE DATA (mm)

Pipe ID	Material	Nom	Sch	O.D.	W.Th	Corr.	Mill	Insul	SG
6"B06	API5L-B	150.00	XS	168.28	10.973	6.000	1.372		0.700

VALVE

Point	Valve Size	Type	Rating	Weight(kg)
AX01	150.00	CATALOG	150#	153.32

FLANGE

Point	Flange Size	Type	Rating	Weight(kg)	Bolt Wt(kg)
AX01	150.00	WELDNECK	150#	11.34	
AX02	150.00	WELDNECK	150#	11.34	
AX07	150.00	WELDNECK	150#	11.34	

SUPPORT DATA (mm, N/mm, N.m/deg)

Supp ID	Type	Conn.to	Stiff.	Gap 1	Gap 2	Friction
AX04 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AX04 1	Guide-V	Ground	Rigid		3.00	0.10
AX05 2	V-stop	Ground	Rigid		2540.00	0.10
AX06 1	Guide-H	Ground	Rigid	3.00	3.00	0.10
AX06 1	Guide-V	Ground	Rigid		3.00	0.10

Gap1: Vstp,Guide-V=down, Lstop,Incl,Tielink=back, Guide-H=Left
Gap2: Vstp,Guide-V=Up , Lstop,Incl,Tielink=Frwd, Guide-H=Right
Gap 3 & 4: Guides (Guide-H) only

SUPPORT LOADS (Max) (N, mm)

Supp ID	Fx	Fy	Fz	Dx	Dy	Dz	Combination
AX04 1					1.97	-5.56	GT1P1(1)
AX04 1		-13031.8	1537.22			7.73	Thermal 2(1)
AX04 1		-15372.2	1537.22			7.73	GT2(1)
AX04 1	234.01			1.81			Wind 1(1)
AX04 1		-2340.38					Gravity(1)
AX05 2		1364.37	162.62			4.54	Thermal 2(1)
AX05 2		-36.81	-697.68			-0.13	Pressure 3(2)
AX05 2		-3506.78	-350.68			-0.04	GT3P3(2)
AX05 2	299.03			2.98			Wind 1(1)
AX05 2		-2990.61					Gravity(1)
AX06 1		-715.27	355.80			1.00	Thermal 2(1)
AX06 1		19.30	-363.74			-0.03	Pressure 3(2)
AX06 1		-3558.05	355.80			1.00	GT2(1)
AX06 1	284.25			0.39			Wind 1(1)
AX06 1		-2842.78					Gravity(1)

For Rotational Supports: Loads are moments (ft-lb) and movements are rotations (deg)

Local1: Up/Down = Vstp, Spr.Hgr, Cons.Hgr, Guides
Frwd/Back = Lstop, Incline, Rotation, Damper, Tielink
Local2: Right/Left =Guides, u-fr/d-fr=Incl&Tlink, Up/Down=Lstop+Z/-Z = Vstp
Local3: Frwd/Back =Guides, r-fr/l-fr=Incl&Tlink, Right/Left=Lstop+X/-X = Vstp
Vstop is Y-vertical shown(Z-Vertical: Local2 = +X/-X, Local3 = +Y/-Y)

ANCHOR LOADS (Max) (N, N.m)

Point	Fx	Fy	Fz	Mx	My	Mz	Combination
AX01		8990.37	6179.47	14779.33			GT2(1)
AX01		11952.75	6144.40	14721.78			Thermal 2(1)
AX01	583.59				281.22	-1502.16	Wind 1(1)
AX07		867.44	-8235.12	647.68			GT2(1)
AX07	341.00				-1084.55	-94.57	Wind 1(1)

REV.	DATE	REVISION	BY	CHKD	APP.

G-Gap
Line Direction for Support Gaps

ANCHOR | LINESTOP | V-STOP | ROTATION | GUIDE | INCLINED | SPRING | CONSTANT | DAMPER

PROJECT NO:	AKABEL	DATE:
ANALYZED BY:	I.O	DATE:
CHECKED BY:	E N	DATE:
APPROVED BY:		DATE:
AREA:		
LINE NO.:		
CLASS:		
FLOW DIAGRAM:		

Bentley

DRG. SIZE D

DRAWING No.

REV.